
Guidance for Industry

Q10 Pharmaceutical Quality System

**U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)**

**April 2009
ICH**

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Guidance for Industry¹

Q10 Pharmaceutical Quality System

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I. INTRODUCTION (1, 1.1)²

This internationally harmonized guidance is intended to assist pharmaceutical manufacturers by describing a model for an effective *quality* management system for the pharmaceutical industry, referred to as the *pharmaceutical quality system*. Throughout this guidance, the term *pharmaceutical quality system* refers to the ICH Q10 model.

ICH Q10 describes one comprehensive model for an effective pharmaceutical quality system that is based on International Organization for Standardization (ISO) quality concepts, includes applicable good manufacturing practice (GMP) regulations, and complements ICH “Q8 Pharmaceutical Development” and ICH “Q9 Quality Risk Management.”³ ICH Q10 is a model for a pharmaceutical quality system that can be implemented throughout the different stages of a product lifecycle. Much of the content of ICH Q10 applicable to manufacturing sites is currently specified by regional GMP requirements. ICH Q10 is not intended to create any new expectations beyond current regulatory requirements. Consequently, the content of ICH Q10 that is additional to current regional GMP requirements is optional.

¹ This guidance was developed within the Expert Working Group (Quality) of the International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) and has been subject to consultation by the regulatory parties, in accordance with the ICH process. This document has been endorsed by the ICH Steering Committee at *Step 4* of the ICH process, June 2008. At *Step 4* of the process, the final draft is recommended for adoption to the regulatory bodies of the European Union, Japan, and the United States.

² Arabic numbers reflect the organizational breakdown of the document endorsed by the ICH Steering Committee at *Step 4* of the ICH process, June 2008.

³ These guidances are available on the Internet at <http://www.fda.gov/cder/guidance/index.htm>. We update guidances periodically. To make sure you have the most recent version of a guidance, check the CDER guidance page at <http://www.fda.gov/cder/guidance/index.htm>

ICH Q10 demonstrates industry and regulatory authorities' support of an effective pharmaceutical quality system to enhance the quality and availability of medicines around the world in the interest of public health. Implementation of ICH Q10 throughout the product lifecycle should facilitate *innovation* and *continual improvement* and strengthen the link between pharmaceutical development and manufacturing activities.

FDA's guidance documents, including this guidance, do not establish legally enforceable responsibilities. Instead, guidances describe the Agency's current thinking on a topic and should be viewed only as recommendations, unless specific regulatory or statutory requirements are cited. The use of the word *should* in Agency guidances means that something is suggested or recommended, but not required.

II. PHARMACEUTICAL QUALITY MANAGEMENT SYSTEM

A. Scope (1.2)

This guidance applies to the systems supporting the development and manufacture of pharmaceutical drug substances (i.e., active pharmaceutical ingredients (APIs)) and drug products, including biotechnology and biological products, throughout the product lifecycle.

The elements of ICH Q10 should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the differences among, and the different goals of, each stage (see section IV (3)).

For the purposes of this guidance, the product lifecycle includes the following technical activities for new and existing products:

- Pharmaceutical Development
 - Drug substance development
 - Formulation development (including container/closure system)
 - Manufacture of investigational products
 - Delivery system development (where relevant)
 - Manufacturing process development and scale-up
 - Analytical method development
- Technology Transfer
 - New product transfers during development through manufacturing
 - Transfers within or between manufacturing and testing sites for marketed products
- Commercial Manufacturing
 - Acquisition and control of materials
 - Provision of facilities, utilities, and equipment
 - Production (including packaging and labeling)
 - Quality control and assurance
 - Release
 - Storage

- Distribution (excluding wholesaler activities)
- Product Discontinuation
 - Retention of documentation
 - Sample retention
 - Continued product assessment and reporting

B. Relationship of ICH Q10 to Regional GMP Requirements, ISO Standards, and ICH Q7 (1.3)

Regional GMP requirements, the ICH guidance “Q7 Good Manufacturing Practice Guidance for Active Pharmaceutical Ingredients,” and ISO quality management system guidelines form the foundation for ICH Q10. To meet the objectives described below, ICH Q10 augments GMPs by describing specific quality system elements and management responsibilities. ICH Q10 provides a harmonized model for a pharmaceutical quality system throughout the lifecycle of a product and is intended to be used together with regional GMP requirements.

The regional GMPs do not explicitly address all stages of the product lifecycle (e.g., development). The quality system elements and management responsibilities described in this guidance are intended to encourage the use of science- and risk-based approaches at each lifecycle stage, thereby promoting continual improvement across the entire product lifecycle.

C. Relationship of ICH Q10 to Regulatory Approaches (1.4)

Regulatory approaches for a specific product or manufacturing facility should be commensurate with the level of product and process understanding, the results of *quality risk management*, and the effectiveness of the pharmaceutical quality system. When implemented, the effectiveness of the pharmaceutical quality system can normally be evaluated during a regulatory inspection at the manufacturing site. Potential opportunities to enhance science- and risk-based regulatory approaches are identified in Annex 1. Regulatory processes will be determined by region.

D. ICH Q10 Objectives (1.5)

Implementation of the Q10 model should result in achievement of three main objectives that complement or enhance regional GMP requirements.

1. Achieve *Product Realization* (1.5.1)

To establish, implement, and maintain a system that allows the delivery of products with the quality attributes appropriate to meet the needs of patients, health care professionals, regulatory authorities (including compliance with approved regulatory filings) and other internal and external customers.

2. Establish and Maintain a *State of Control* (1.5.2)

To develop and use effective monitoring and control systems for process performance and product quality, thereby providing assurance of continued suitability and *capability of processes*. Quality risk management can be useful in identifying the monitoring and control systems.

3 *Facilitate Continual Improvement (1.5.3)*

To identify and implement appropriate product quality improvements, process improvements, variability reduction, innovations, and pharmaceutical quality system enhancements, thereby increasing the ability to fulfill a pharmaceutical manufacturer's own quality needs consistently. Quality risk management can be useful for identifying and prioritizing areas for continual improvement.

E. *Enablers: Knowledge Management and Quality Risk Management (1.6)*

Use of *knowledge management* and quality risk management will enable a company to implement ICH Q10 effectively and successfully. These enablers will facilitate achievement of the objectives described in section II.D (1.5) above by providing the means for science- and risk-based decisions related to product quality.

1. *Knowledge Management (1.6.1)*

Product and process knowledge should be managed from development through the commercial life of the product up to and including product discontinuation. For example, development activities using scientific approaches provide knowledge for product and process understanding. Knowledge management is a systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components. Sources of knowledge include, but are not limited to, prior knowledge (public domain or internally documented); pharmaceutical development studies; technology transfer activities; process validation studies over the product lifecycle; manufacturing experience; innovation; continual improvement; and *change management* activities.

2. *Quality Risk Management (1.6.2)*

Quality risk management is integral to an effective pharmaceutical quality system. It can provide a proactive approach to identifying, scientifically evaluating, and controlling potential risks to quality. It facilitates continual improvement of process performance and product quality throughout the product lifecycle. ICH Q9 provides principles and examples of tools for quality risk management that can be applied to different aspects of pharmaceutical quality.

F. *Design and Content Considerations (1.7)*

- (a) The design, organization, and documentation of the pharmaceutical quality system should be well structured and clear to facilitate common understanding and consistent application.
- (b) The elements of ICH Q10 should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the different goals

and knowledge available for each stage.

- (c) The size and complexity of the company's activities should be taken into consideration when developing a new pharmaceutical quality system or modifying an existing one. The design of the pharmaceutical quality system should incorporate appropriate risk management principles. While some aspects of the pharmaceutical quality system can be company wide and others site specific, the effectiveness of the pharmaceutical quality system is normally demonstrated at the site level.
- (d) The pharmaceutical quality system should include appropriate processes, resources, and responsibilities to provide assurance of the quality of *outsourced activities* and purchased materials as described in section III.G (2.7).
- (e) Management responsibilities, as described in section III (2), should be identified within the pharmaceutical quality system.
- (f) The pharmaceutical quality system should include the following elements, as described in section IV (3): process performance and product quality monitoring, *corrective* and *preventive action*, change management, and management review.
- (g) *Performance indicators*, as described in section V (4), should be identified and used to monitor the effectiveness of processes within the pharmaceutical quality system.

G. Quality Manual (1.8)

A *Quality Manual* or equivalent documentation approach should be established and should contain the description of the pharmaceutical quality system. The description should include:

- (a) The *quality policy* (see section III (2)).
- (b) The scope of the pharmaceutical quality system.
- (c) Identification of the pharmaceutical quality system processes, as well as their sequences, linkages, and interdependencies. Process maps and flow charts can be useful tools to facilitate depicting pharmaceutical quality system processes in a visual manner.
- (d) Management responsibilities within the pharmaceutical quality system (see section III (2)).

III. MANAGEMENT RESPONSIBILITY (2)

Leadership is essential to establish and maintain a company-wide commitment to quality and for the performance of the pharmaceutical quality system.

A. Management Commitment (2.1)

- (a) *Senior management* has the ultimate responsibility to ensure an effective pharmaceutical quality system is in place to achieve the *quality objectives*, and that

roles, responsibilities, and authorities are defined, communicated, and implemented throughout the company.

(b) Management should:

- (1) Participate in the design, implementation, monitoring, and maintenance of an effective pharmaceutical quality system.
- (2) Demonstrate strong and visible support for the pharmaceutical quality system and ensure its implementation throughout their organization.
- (3) Ensure a timely and effective communication and escalation process exists to raise quality issues to the appropriate levels of management.
- (4) Define individual and collective roles, responsibilities, authorities, and inter-relationships of all organizational units related to the pharmaceutical quality system. Ensure these interactions are communicated and understood at all levels of the organization. An independent quality unit/structure with authority to fulfill certain pharmaceutical quality system responsibilities is required by regional regulations.
- (5) Conduct management reviews of process performance and product quality and of the pharmaceutical quality system.
- (6) Advocate continual improvement.
- (7) Commit appropriate resources.

B. Quality Policy (2.2)

- (a) Senior management should establish a quality policy that describes the overall intentions and direction of the company related to quality.
- (b) The quality policy should include an expectation to comply with applicable regulatory requirements and should facilitate continual improvement of the pharmaceutical quality system.
- (c) The quality policy should be communicated to and understood by personnel at all levels in the company.
- (d) The quality policy should be reviewed periodically for continuing effectiveness.

C. Quality Planning (2.3)

- (a) Senior management should ensure the quality objectives to implement the quality policy are defined and communicated.
- (b) Quality objectives should be supported by all relevant levels of the company.
- (c) Quality objectives should align with the company's strategies and be consistent with the quality policy.
- (d) Management should provide the appropriate resources and training to achieve the

quality objectives.

- (e) Performance indicators that measure progress against quality objectives should be established, monitored, communicated regularly, and acted upon as appropriate as described in section V.A (4.1) of this document.

D. Resource Management (2.4)

- (a) Management should determine and provide adequate and appropriate resources (human, financial, materials, facilities, and equipment) to implement and maintain the pharmaceutical quality system and continually improve its effectiveness.
- (b) Management should ensure that resources are appropriately applied to a specific product, process, or site.

E. Internal Communication (2.5)

- (a) Management should ensure appropriate communication processes are established and implemented within the organization.
- (b) Communications processes should ensure the flow of appropriate information between all levels of the company.
- (c) Communication processes should ensure the appropriate and timely escalation of certain product quality and pharmaceutical quality system issues.

F. Management Review (2.6)

- (a) Senior management should be responsible for pharmaceutical quality system governance through management review to ensure its continuing suitability and effectiveness.
- (b) Management should assess the conclusions of periodic reviews of process performance and product quality and of the pharmaceutical quality system, as described in sections IV (3) and V (4).

G. Management of Outsourced Activities and Purchased Materials (2.7)

The pharmaceutical quality system, including the management responsibilities described in this section, extends to the control and review of any outsourced activities and quality of purchased materials. The pharmaceutical company is ultimately responsible to ensure processes are in place to assure the control of outsourced activities and quality of purchased materials. These processes should incorporate quality risk management and include:

- (a) Assessing prior to outsourcing operations or selecting material suppliers, the suitability and competence of the other party to carry out the activity or provide the material

using a defined supply chain (e.g., audits, material evaluations, qualification).

- (b) Defining the responsibilities and communication processes for quality-related activities of the involved parties. For outsourced activities, this should be included in a written agreement between the contract giver and contract acceptor.
- (c) Monitoring and review of the performance of the contract acceptor or the quality of the material from the provider, and the identification and implementation of any essential improvements.
- (d) Monitoring incoming ingredients and materials to ensure they are from approved sources using the agreed supply chain.

H. Management of Change in Product Ownership (2.8)

When product ownership changes (e.g., through acquisitions), management should consider the complexity of this and ensure:

- (a) The ongoing responsibilities are defined for each company involved
- (b) The essential information is transferred

IV. CONTINUAL IMPROVEMENT OF PROCESS PERFORMANCE AND PRODUCT QUALITY (3)

This section describes the lifecycle stage goals and the four specific pharmaceutical quality system elements that augment regional requirements to achieve the ICH Q10 objectives, as defined in section II.D (1.5). It does not restate all regional GMP requirements.

A. Lifecycle Stage Goals (3.1)

The goals of each product lifecycle stage are described below.

1. *Pharmaceutical Development (3.1.1)*

The goal of pharmaceutical development activities is to design a product and its manufacturing process to consistently deliver the intended performance and meet the needs of patients and healthcare professionals, and regulatory authorities and internal customers' requirements. Approaches to pharmaceutical development are described in ICH Q8. The results of exploratory and clinical development studies, while outside the scope of this guidance, are inputs to pharmaceutical development.

2. *Technology Transfer (3.1.2)*

The goal of technology transfer activities is to transfer product and process knowledge between development and manufacturing, and within or between manufacturing sites to achieve product

realization. This knowledge forms the basis for the manufacturing process, *control strategy*, process validation approach, and ongoing continual improvement.

3. *Commercial Manufacturing (3.1.3)*

The goals of manufacturing activities include achieving product realization, establishing and maintaining a state of control, and facilitating continual improvement. The pharmaceutical quality system should assure that the desired product quality is routinely met, suitable process performance is achieved, the set of controls are appropriate, improvement opportunities are identified and evaluated, and the body of knowledge is continually expanded.

4. *Product Discontinuation (3.1.4)*

The goal of product discontinuation activities is to manage the terminal stage of the product lifecycle effectively. For product discontinuation, a predefined approach should be used to manage activities such as retention of documentation and samples and continued product assessment (e.g., complaint handling and stability) and reporting in accordance with regulatory requirements.

B. Pharmaceutical Quality System Elements (3.2)

The elements described below might be required in part under regional GMP regulations. However, the Q10 model's intent is to enhance these elements to promote the lifecycle approach to product quality. These four elements are:

- Process performance and product quality monitoring system
- *Corrective action* and *preventive action* (CAPA) system
- Change management system
- Management review of process performance and product quality

These elements should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the differences among the stages and the different goals of each stage. Throughout the product lifecycle, companies are encouraged to evaluate opportunities for innovative approaches to improve product quality.

Each element is followed by a table of example applications of the element to the stages of the pharmaceutical lifecycle.

1. *Process Performance and Product Quality Monitoring System 3.2.1*

Pharmaceutical companies should plan and execute a system for the monitoring of process performance and product quality to ensure a state of control is maintained. An effective monitoring system provides assurance of the continued capability of processes and controls to produce a product of desired quality and to identify areas for continual improvement. The process performance and product quality monitoring system should:

- (a) Use quality risk management to establish the control strategy. This can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control. The control strategy should facilitate timely *feedback/feedforward* and appropriate corrective action and preventive action.
- (b) Provide the tools for measurement and analysis of parameters and attributes identified in the control strategy (e.g., data management and statistical tools).
- (c) Analyze parameters and attributes identified in the control strategy to verify continued operation within a state of control.
- (d) Identify sources of variation affecting process performance and product quality for potential continual improvement activities to reduce or control variation.
- (e) Include feedback on product quality from both internal and external sources (e.g., complaints, product rejections, nonconformances, recalls, deviations, audits and regulatory inspections, and findings).
- (f) Provide knowledge to enhance process understanding, enrich the *design space* (where established), and enable innovative approaches to process validation.

Table I: Application of Process Performance and Product Quality Monitoring System Throughout the Product Lifecycle

Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Process and product knowledge generated and process and product monitoring conducted throughout development can be used to establish a control strategy for manufacturing.	Monitoring during scale-up activities can provide a preliminary indication of process performance and the successful integration into manufacturing. Knowledge obtained during transfer and scale-up activities can be useful in further developing the control strategy.	A well-defined system for process performance and product quality monitoring should be applied to assure performance within a state of control and to identify improvement areas.	Once manufacturing ceases, monitoring such as stability testing should continue to completion of the studies. Appropriate action on marketed product should continue to be executed according to regional regulations.

2. Corrective Action and Preventive Action (CAPA) System (3.2.2)

The pharmaceutical company should have a system for implementing corrective actions and preventive actions resulting from the investigation of complaints, product rejections, nonconformances, recalls, deviations, audits, regulatory inspections and findings, and trends from process performance and product quality monitoring. A structured approach to the

investigation process should be used with the objective of determining the root cause. The level of effort, formality, and documentation of the investigation should be commensurate with the level of risk, in line with ICH Q9. CAPA methodology should result in product and process improvements and enhanced product and process understanding.

Table II: Application of Corrective Action and Preventive Action System Throughout the Product Lifecycle

Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Product or process variability is explored. CAPA methodology is useful where corrective actions and preventive actions are incorporated into the iterative design and development process.	CAPA can be used as an effective system for feedback, feedforward, and continual improvement.	CAPA should be used, and the effectiveness of the actions should be evaluated.	CAPA should continue after the product is discontinued. The impact on product remaining on the market should be considered, as well as other products that might be affected.

3. *Change Management System (3.2.3)*

Innovation, continual improvement, the outputs of process performance and product quality monitoring, and CAPA drive change. To evaluate, approve, and implement these changes properly, a company should have an effective change management system. There is generally a difference in formality of change management processes prior to the initial regulatory submission and after submission, where changes to the regulatory filing might be required under regional requirements.

The change management system ensures continual improvement is undertaken in a timely and effective manner. It should provide a high degree of assurance there are no unintended consequences of the change.

The change management system should include the following, as appropriate for the stage of the lifecycle:

- (a) Quality risk management should be utilized to evaluate proposed changes. The level of effort and formality of the evaluation should be commensurate with the level of risk.
- (b) Proposed changes should be evaluated relative to the marketing authorization, including design space, where established, and/or current product and process understanding. There should be an assessment to determine whether a change to the regulatory filing is required under regional requirements. As stated in ICH Q8, working within the design space is not considered a change (from a regulatory filing perspective). However, from a pharmaceutical quality system standpoint, all changes should be evaluated by a company's change management system.
- (c) Proposed changes should be evaluated by expert teams contributing the appropriate

expertise and knowledge from relevant areas (e.g., Pharmaceutical Development, Manufacturing, Quality, Regulatory Affairs, and Medical) to ensure the change is technically justified. Prospective evaluation criteria for a proposed change should be set.

- (d) After implementation, an evaluation of the change should be undertaken to confirm the change objectives were achieved and that there was no deleterious impact on product quality.

Table III: Application of Change Management System Throughout the Product Lifecycle

Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Change is an inherent part of the development process and should be documented; the formality of the change management process should be consistent with the stage of pharmaceutical development.	The change management system should provide management and documentation of adjustments made to the process during technology transfer activities.	A formal change management system should be in place for commercial manufacturing. Oversight by the quality unit should provide assurance of appropriate science- and risk-based assessments.	Any changes after product discontinuation should go through an appropriate change management system.

4. Management Review of Process Performance and Product Quality (3.2.4)

Management review should provide assurance that process performance and product quality are managed over the lifecycle. Depending on the size and complexity of the company, management review can be a series of reviews at various levels of management and should include a timely and effective communication and escalation process to raise appropriate quality issues to senior levels of management for review.

- (a) The management review system should include:
- (1) The results of regulatory inspections and findings, audits and other assessments, and commitments made to regulatory authorities
 - (2) Periodic quality reviews, that can include:
 - (i) Measures of customer satisfaction such as product quality complaints and recalls
 - (ii) Conclusions of process performance and product quality monitoring
 - (iii) The effectiveness of process and product changes including those arising from corrective action and preventive actions
 - (3) Any follow-up actions from previous management reviews
- (b) The management review system should identify appropriate actions, such as:
- (1) Improvements to manufacturing processes and products
 - (2) Provision, training, and/or realignment of resources

(3) Capture and dissemination of knowledge

Table IV: Application of Management Review of Process Performance and Product Quality Throughout the Product Lifecycle

Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Aspects of management review can be performed to ensure adequacy of the product and process design.	Aspects of management review should be performed to ensure the developed product and process can be manufactured at commercial scale.	Management review should be a structured system, as described above, and should support continual improvement.	Management review should include such items as product stability and product quality complaints.

V. CONTINUAL IMPROVEMENT OF THE PHARMACEUTICAL QUALITY SYSTEM (4)

This section describes activities that should be conducted to manage and continually improve the pharmaceutical quality system.

A. Management Review of the Pharmaceutical Quality System (4.1)

Management should have a formal process for reviewing the pharmaceutical quality system on a periodic basis. The review should include:

- (a) Measurement of achievement of pharmaceutical quality system objectives
- (b) Assessment of performance indicators that can be used to monitor the effectiveness of processes within the pharmaceutical quality system, such as:
 - (1) Complaint, deviation, CAPA and change management processes
 - (2) Feedback on outsourced activities
 - (3) Self-assessment processes including risk assessments, trending, and audits
 - (4) External assessments such as regulatory inspections and findings and customer audits

B. Monitoring of Internal and External Factors That Can Have an Impact on the Pharmaceutical Quality System (4.2)

Factors monitored by management can include:

- (a) Emerging regulations, guidance, and quality issues that can have an impact on the Pharmaceutical Quality System
- (b) Innovations that might enhance the pharmaceutical quality system
- (c) Changes in business environment and objectives

- (d) Changes in product ownership

C. Outcomes of Management Review and Monitoring (4.3)

The outcome of management review of the pharmaceutical quality system and monitoring of internal and external factors can include:

- (a) Improvements to the pharmaceutical quality system and related processes
- (b) Allocation or reallocation of resources and/or personnel training
- (c) Revisions to quality policy and quality objectives
- (d) Documentation and timely and effective communication of the results of the management review and actions, including escalation of appropriate issues to senior management

VI. GLOSSARY (5)

ICH and ISO definitions are used in ICH Q10 where they exist. For the purpose of ICH Q10, where the words “requirement”, “requirements,” or “necessary” appear in an ISO definition, they do not necessarily reflect a regulatory requirement. The source of the definition is identified in parentheses after the definition. Where no appropriate ICH or ISO definition was available, an ICH Q10 definition was developed.

Capability of a Process: Ability of a process to realize a product that will fulfill the requirements of that product. The concept of process capability can also be defined in statistical terms. (ISO 9000:2005)

Change Management: A systematic approach to proposing, evaluating, approving, implementing, and reviewing changes. (ICH Q10)

Continual Improvement: Recurring activity to increase the ability to fulfill requirements. (ISO 9000:2005)

Control Strategy: A planned set of controls, derived from current product and process understanding, that assures process performance and product quality. The controls can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control. (ICH Q10)

Corrective Action: Action to eliminate the cause of a detected nonconformity or other undesirable situation. NOTE: Corrective action is taken to prevent recurrence whereas preventive action is taken to prevent occurrence. (ISO 9000:2005)

Design Space: The multidimensional combination and interaction of input variables (e.g., material attributes) and process parameters that have been demonstrated to provide assurance of quality. (ICH Q8)

Enabler: A tool or process that provides the means to achieve an objective. (ICH Q10)

Feedback/Feedforward:

Feedback: The modification or control of a process or system by its results or effects.

Feedforward: The modification or control of a process using its anticipated results or effects. (Oxford Dictionary of English by Oxford University Press, 2003)

Feedback/feedforward can be applied technically in process control strategies and conceptually in quality management. (ICH Q10)

Innovation: The introduction of new technologies or methodologies. (ICH Q10)

Knowledge Management: Systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components. (ICH Q10)

Outsourced Activities: Activities conducted by a contract acceptor under a written agreement with a contract giver. (ICH Q10)

Performance Indicators: Measurable values used to quantify quality objectives to reflect the performance of an organization, process, or system, also known as *performance metrics* in some regions. (ICH Q10)

Pharmaceutical Quality System (PQS): Management system to direct and control a pharmaceutical company with regard to quality. (ICH Q10 based upon ISO 9000:2005)

Preventive Action: Action to eliminate the cause of a potential nonconformity or other undesirable potential situation. NOTE: Preventive action is taken to prevent occurrence whereas corrective action is taken to prevent recurrence. (ISO 9000:2005)

Product Realization: Achievement of a product with the quality attributes appropriate to meet the needs of patients, health care professionals, and regulatory authorities (including compliance with marketing authorization) and internal customers requirements. (ICH Q10)

Quality: The degree to which a set of inherent properties of a product, system, or process fulfils requirements. (ICH Q9)

Quality Manual: Document specifying the quality management system of an organization. (ISO 9000:2005)

Quality Objectives: A means to translate the quality policy and strategies into measurable activities. (ICH Q10)

Quality Planning: Part of quality management focused on setting quality objectives and specifying necessary operational processes and related resources to fulfill the quality objectives. (ISO 9000:2005)

Quality Policy: Overall intentions and direction of an organization related to quality as formally expressed by senior management. (ISO 9000:2005)

Quality Risk Management: A systematic process for the assessment, control, communication, and review of risks to the quality of the drug (medicinal) product across the product lifecycle. (ICH Q9)

Senior Management: Person(s) who direct and control a company or site at the highest levels with the authority and responsibility to mobilize resources within the company or site. (ICH Q10 based in part on ISO 9000:2005)

State of Control: A condition in which the set of controls consistently provides assurance of continued process performance and product quality. (ICH Q10)

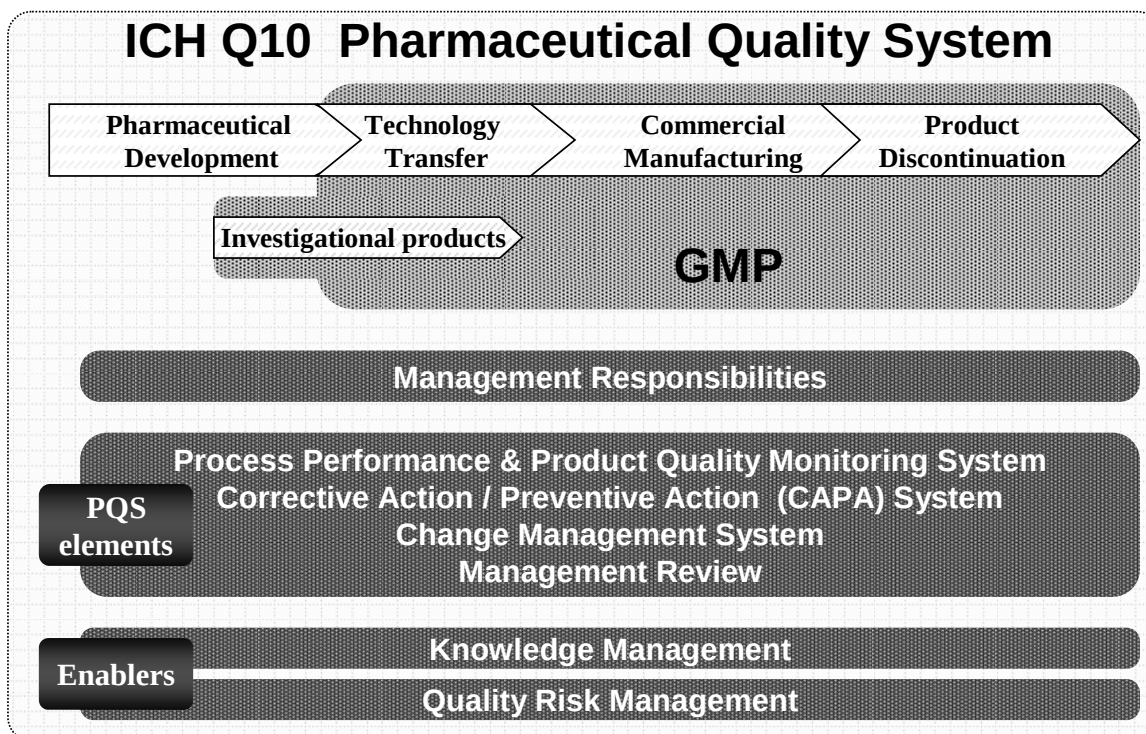
Annex I: Potential Opportunities To Enhance Science- and Risk-Based Regulatory Approaches *

*Note: This annex reflects potential opportunities to enhance regulatory approaches. The actual regulatory process will be determined by region.

Scenario	Potential Opportunity
1. Comply with GMPs	Compliance – status quo
2. Demonstrate effective pharmaceutical quality system, including effective use of quality risk management principles (e.g., ICH Q9 and ICH Q10).	Opportunity to: • increase use of risk-based approaches for regulatory inspections
3. Demonstrate product and process understanding, including effective use of quality risk management principles (e.g., ICH Q8 and ICH Q9).	Opportunity to: • facilitate science-based pharmaceutical quality assessment • enable innovative approaches to process validation • establish real-time release mechanisms
4. Demonstrate effective pharmaceutical quality system and product and process understanding, including the use of quality risk management principles (e.g., ICH Q8, ICH Q9, and ICH Q10).	Opportunity to: • increase use of risk-based approaches for regulatory inspections; • facilitate science-based pharmaceutical quality assessment; • optimize science- and risk-based postapproval change processes to maximize benefits from innovation and continual improvement; • enable innovative approaches to process validation; • establish real-time release mechanisms.

Annex 2

Diagram of the ICH Q10 Pharmaceutical Quality System Model



This diagram illustrates the major features of the ICH Q10 Pharmaceutical Quality System (PQS) model. The PQS covers the entire lifecycle of a product including pharmaceutical development, technology transfer, commercial manufacturing, and product discontinuation as illustrated by the upper portion of the diagram. The PQS augments regional GMPs as illustrated in the diagram. The diagram also illustrates that regional GMPs apply to the manufacture of investigational products.

The next horizontal bar illustrates the importance of management responsibilities explained in section III (2) to all stages of the product lifecycle. The following horizontal bar lists the PQS elements that serve as the major pillars under the PQS model. These elements should be applied appropriately and proportionally to each lifecycle stage, recognizing opportunities to identify areas for continual improvement.

The bottom set of horizontal bars illustrates the enablers: knowledge management and quality risk management, which are applicable throughout the lifecycle stages. These enablers support the PQS goals of achieving product realization, establishing and maintaining a state of control, and facilitating continual improvement.

1 Chapter 4: Documentation

2 **Reasons for changes:** The GMP/GDP Inspectors Working Group and the PIC/S Committee jointly
3 recommended that the current version of Chapter 4, on documentation, is revised to reflect changes in
4 regulatory and manufacturing environments. The revised guideline clarifies requirements and expecta-
5 tions from Regulatory Authorities with regards to documentation and takes into account related changes
6 for Annex 11 of the GMP Guide.

Document map

Principle

Data governance systems

Risk management

General requirements for documentation

Master Documents

Generation and Control of Documentation

Good documentation practice

Signatures in GMP relevant documentation

Retention of documents

Data Integrity in documentation

Hybrid Systems

Glossary

7 PRINCIPLE

- 8 4.1. Documentation constitutes an essential part of the quality assurance system and is key to op-
9 erating in compliance with GMP requirements. The various types of documents and means
10 used should be fully understood and defined in the regulated user's pharmaceutical quality
11 system.
- 12 4.2. It should be determined by the regulated user which legal provisions apply to documentation
13 considering new technologies, hybrid solutions and services used.
- 14 4.3. Appropriate documentation practices should be applied with respect to the type of document
15 regardless of the applied technology or service used.
- 16 4.4. The present document was supplemented by requirements regarding new technologies, hybrid
17 solutions and new services, whereby a risk-based approach as element of a data governance
18 system is considered pivotal for scalability of integrity control measures.
- 19 4.5. Quality risk management principles should be applied to ensure that the pharmaceuti-
20 cal quality system includes sufficient instructional details. It should facilitate a common
21 understanding of the requirements. In addition to providing for recording of the various pro-
22 cesses and risk-based evaluation of any observations, it should demonstrate the ongoing appli-
23 cation of all requirements.
- 24 4.6. Suitable controls should be implemented using a risk-based approach to ensure the accuracy,
25 integrity, availability, and legibility of documents. Documents should be free from errors and
26 available in human readable form.
- 27 4.7. Documentation may exist in a variety of forms, including paper-based, electronic, or other
28 means (e.g. photography, imagery, video and audio recordings). The main objective of the
29 documentation system is to establish, control, monitor and record all activities which directly
30 or indirectly impact on all aspects of the quality and safety of medicinal products, using a risk-
31 based approach.
- 32 4.8. Whether documents are created, stored, and managed electronically, paper based, by other
33 means or through a hybrid system, they must meet the same GMP requirements for legibility,
34 accuracy, integrity, and completeness throughout the whole lifecycle. This also applies when
35 documentation is outsourced.
- 36 4.9. There are two primary types of documentation used to manage and demonstrate GMP compli-
37 ance:
 - 38 i. instructions (directions, requirements) and
 - 39 ii. records/reports.

40 DATA GOVERNANCE SYSTEMS

- 41 4.10. Regardless of how documents are created, handled, stored, and managed (i.e., using elec-
42 tronic, paper-based, or hybrid systems), the regulated user should establish a data governance
43 system integral to the pharmaceutical quality system to define, prioritise and communicate
44 their data integrity risk management activities. Arrangements for data governance should be
45 documented and reviewed regularly.
- 46 4.11. Regulated users should design and operate a data governance system which provides an ac-
47 ceptable state of control based on the risk assessment, which is documented with supporting
48 rationale. A data governance system should be consistent with the principles of quality risk
49 management.
- 50 4.12. To ensure integrity of data the governance system should cover the entire data lifecycle and
51 ensure controls commensurate with the principle of quality risk management. The data lifecy-
52 cle should refer to:
 - 53 i. Creation and recording of data.
 - 54 ii. Processing of (raw) data to reported (derived) data.

- iii. Verification of completeness, consistency and accuracy of all data (raw and derived data). For derived data the traceability which allows reconstruction of all data processing activities should be maintained.
 - iv. Decision making relying on data (or derived data).
 - v. Retaining, archiving and retrieval of data. To protect it from loss or unauthorised alteration it should be commensurate with the principles of quality risk management.
 - vi. Retirement or destruction of data at the end of the lifecycle in a controlled manner.
- 4.13. Data governance systems should rely on a risk management approach and consider:
- i. Data criticality (impact to decision making and product quality) and
 - ii. Data risk (opportunity for data alteration and deletion, and likelihood of detection / visibility of changes by the regulated user's routine review processes).
- 4.14. Data governance systems should rely on the incorporation of suitably designed systems, the use of technologies and data security measures, combined with specific expertise to ensure that data integrity is effectively controlled over the data lifecycle.
- 4.15. Data governance systems should address data ownership throughout the entire lifecycle.
- 4.16. Data governance systems should consider the design, operation and monitoring of processes and systems to comply with the principles of data integrity.
- 4.17. Data governance systems should consider risk mitigation. The effectiveness of risk mitigation measures should be reviewed regularly, regardless of whether they are temporary or permanent. Residual risks should be reviewed periodically and communicated to management.
- 4.18. Data governance systems should consider and ensure the periodic review of service provider's data management policies and risk control strategies intended to minimise potential risks to data integrity. The frequency of such reviews should be based on the criticality of the services provided, using risk management principles.

RISK MANAGEMENT

- 4.19. The regulated user should adopt a risk-based approach in documentation throughout the entire lifecycle of data, regardless of the technology, hybrid solution or service used and should demonstrate an understanding for data criticality, data risk and data quality.
- 4.20. Controls over the data lifecycle should be established which are commensurate with the principles of quality risk management. The depth of data governance and risk management activities should be justified and commensurate with the risks to product quality and patient safety.
- 4.21. Decisions on the extent of measures to ensure data integrity should be based on a documented rationale and documented risk assessment taking into consideration data criticality and data risk.
- 4.22. Irrespective of processes used to generate electronic data, they must be included in the requirements for the qualification or validation of the relevant computerised systems according to Annex 11.

GENERAL REQUIREMENTS FOR DOCUMENTATION

- 4.23. The pharmaceutical quality system should describe all documents required to ensure product quality and patient safety. Documents may be created, recorded, provided, approved, communicated, stored, and archived electronically, paper based or in a hybrid system. The reliance on electronic, paper-based or different means, hybrid solutions or hosted services in maintenance and retention of documentation requires the compliance with all EU GMP provisions including Annex 11 if decision making in manufacturing (e.g. batch release based on in-process controls and process analytical technologies) is supported by automatic validation scripts or artificial intelligence (Annex 22).
- 4.24. The accountability for the integrity of documents, records or (raw) data produced or processed

with artificial intelligence or any other automatic means (e.g. validation scripts) rests with the regulated user.

4.25. The support by any automatic means (e.g. validation scripts or artificial intelligence) should be included in a pharmaceutical quality system regardless of the service located on premise or as a hosted service. The records created electronically should enable a trend analysis of quality-critical data.

4.26. To ensure data integrity, data which is recorded or processed electronically should not be converted to or stored in a paper form unless it meets the requirements set out in section 13 “hybrid systems” or the conversion is validated or verified for accuracy.

MASTER DOCUMENTS

4.27. Specifically required master documents (not exhaustive list):

i. **Site Master File:** Refer to EU GMP Guidelines, Volume 4 “Explanatory Notes on the preparation of a Site Master File”.

ii. **Validation Master Plan:** A document describing the key elements of the site qualification and validation program. Master documents should be evaluated and reviewed on a regular basis.

iii. **Instructions** (directions, or requirements) type:

- **Specification:** Refer to glossary for definition

- **Manufacturing Formulae, Processing, Packaging and Testing Instruction:** Provide complete detail on all the starting materials, equipment, and computerised systems (if any) to be used and specify all processing, packaging, sampling, and testing instructions to ensure batch to batch consistency. In-process controls and process analytical technologies to be employed should be specified where relevant, together with acceptance criteria.

- **Procedures:** (Otherwise known as Standard Operating Procedures, or SOPs), documented set of instructions for performing and recording operations.

- **Protocol:** defined set of activities to provide instructions for performing and recording certain discreet operations.

- **Technical / Quality Agreement:** Written proof of agreement between contract givers and acceptors for outsourced activities.

iv. **Record/Report type:**

- **Record:** Provide evidence of various actions taken to demonstrate compliance with instructions, e.g. activities, events, investigations, and in the case of manufactured batches a history of each batch of product, including its distribution. Records include the raw data which is used to generate other records. For electronic records regulated users should define which data are to be used as raw data. At least, all data on which quality decisions are based should be defined as raw data.

Records can also exist as hybrid records which is a combination of paper records, electronic records or by other means. The completeness and integrity of records, including all relevant raw data and meta data should be ensured and protected based on risk.

- **Certificate of Analysis:** Provide a summary of testing results on samples of products or materials¹ together with the evaluation for compliance to a stated specification.

¹ Alternatively, the certification may be based, in-whole or in-part, on the assessment of real time data (summaries and exception reports) from batch related process analytical technology (PAT), parameters or metrics as per the approved marketing authorisation dossier.

- 145 - **Report:** Document the conduct of exercises, studies, assessments, projects or inves-
146 tigations, together with results, conclusions and recommendations.

147 **Specifications**

- 148 4.28. There should be approved and updated specifications for starting and packaging materials,
149 intermediate, bulk, finished products, process aids and other quality critical material, as ap-
150 plicable. Specifications should include all attributes which are relevant for product quality on
151 each stage of material or manufacture.

152 **Specifications for starting and packaging materials**

- 153 4.29. Specifications for starting and primary or printed packaging materials should include the prod-
154 uct and its reference code, if applicable:
- 155 i. A description of the materials, including:
 - 156 • The designated name and the internal code reference.
 - 157 • The reference, if any, to a pharmacopeial monograph.
 - 158 • The approved suppliers and, if applicable, the original producer of the material.
 - 159 • A specimen of printed materials.
 - 160 ii. Directions for sampling and testing.
 - 161 iii. Qualitative and quantitative requirements with acceptance limits.
 - 162 iv. Storage conditions and precautions.
 - 163 v. The maximum period of storage before re-examination.

164 **Specifications for intermediate and bulk products**

- 165 4.30. Specifications for intermediate and bulk products should be available for critical steps or if
166 these are purchased or dispatched. The specifications should be similar to specifications for
167 starting materials or for finished products, as appropriate.

168 **Specifications for finished products**

- 169 4.31. Specifications for finished products should include or provide reference to:
- 170 vi. The designated name of the product and the code reference where applicable.
 - 171 vii. The formula.
 - 172 viii. A description of the pharmaceutical form and package details.
 - 173 ix. Directions for sampling and testing.
 - 174 x. The qualitative and quantitative requirements, with the acceptance limits.
 - 175 xi. The storage conditions and any special handling precautions, where applicable.
 - 176 xii. The shelf-life.
- 177

178 **Manufacturing Formula and Processing Instructions**

- 179 4.32. Approved, written Manufacturing Formula and Processing Instructions should exist for each
180 product and batch size to be manufactured.

- 181 4.33. The Manufacturing Formula should include:

- 182 i. The name of the product, with a product reference code relating to its specification.
- 183 ii. A description of the pharmaceutical form, strength of the product and batch size.
- 184 iii. A list of all starting materials to be used, with the amount of each, described;

185 mention should be made of any substance that may disappear while processing, of
186 processing aids needed or any other material relevant for product quality.

187 iv. A statement of the expected final yield with the acceptable limits, and of relevant in-
188 termediate yields, where applicable.

189 4.34. The Processing Instructions should include:

190 i. A statement of the processing location and the principal equipment to be used.

191 ii. The methods, or reference to the methods, to be used for preparing the critical equip-
192 ment (e.g. cleaning, assembling, calibrating, sterilising).

193 iii. Checks that the equipment and workstation are clear of previous products, docu-
194 ments or materials not required for the planned process, and that equipment is clean
195 and suitable for use.

196 iv. Detailed stepwise processing instructions [e.g. checks on materials, pre-treatments,
197 sequence for adding materials, critical process parameters (time, temp etc)].

198 v. The instructions for any in-process controls with their limits.

199 vi. Where necessary, the requirements for bulk storage of the products; including the
200 container, labelling and special storage conditions where applicable.

201 vii. Any special precautions to be observed.

202 **Packaging Instructions**

203 4.35. Approved Packaging Instructions for each product, pack size and type should exist. These
204 should include, or have a reference to, the following:

205 i. Name of the product; including the batch number of bulk and finished product.

206 ii. Description of its pharmaceutical form, and strength where applicable.

207 iii. The pack size expressed in terms of the number, weight or volume of the product in
208 the final container.

209 iv. A complete list of all the packaging materials required, including quantities, sizes
210 and types, with the code or reference number relating to the specifications of each
211 packaging material.

212 v. Where appropriate, an example or reproduction of the relevant printed packaging
213 materials, and specimens indicating where to apply batch number references, and
214 shelf life of the product.

215 vi. Checks that the equipment and workstation are clear of previous products, docu-
216 ments or materials not required for the planned packaging operations (line clear-
217 ance), and that equipment is clean and suitable for use.

218 vii. Checks on functioning of any electronic code readers, label counters or similar de-
219 vices.

220 viii. Special precautions to be observed, including a careful examination of the area and
221 equipment in order to ascertain the line clearance before operations begin.

222 ix. A description of the packaging operation, including any significant subsidiary opera-
223 tions, and equipment to be used.

224 x. Details of in-process controls with instructions for sampling and acceptance limits.

225 **Batch Processing Record**

226 4.36. A Batch Processing Record should be kept for each batch processed. It should be based on

the relevant parts of the currently approved Manufacturing Formula and Processing Instructions, and should contain the following information:

- i. The name and batch number of the product.
- ii. Dates and times of commencement, of significant intermediate stages and of completion of production.
- iii. Identification (initials) of the operator(s) who performed each significant step of the process and, where appropriate, identification (initials) of the person who checked these operations.
- iv. The batch number and/or analytical control number as well as the quantities of each starting material weighed (including the batch number and amount of any recovered or reprocessed material added).
- v. Any relevant processing operation or event and major equipment used.
- vi. A record of the in-process controls and the initials of the person(s) carrying them out, and the results obtained.
- vii. The product yield obtained at different and pertinent stages of manufacture.
- viii. Notes on special problems including details, with signed authorisation for any deviation from the Manufacturing Formula and Processing Instructions.
- ix. Approval by the person responsible for the processing operations.

Note: Where a validated process is continuously monitored and controlled, then automatically generated reports may be limited to compliance summaries and exception/ out-of-specification (OOS) data reports.

With regards to decision making in manufacturing supported by automatic validation scripts or artificial intelligence refer to paragraph 4 of this document.

Batch Packaging Record

4.37. A Batch Packaging Record should be kept for each batch or part batch processed. It should be based on the relevant parts of the Packaging Instructions.

4.38. The batch packaging record should contain the following information:

- i. The name and batch number of the product.
- ii. The date(s) and times of the packaging operations.
- iii. Identification (initials) of the operator(s) who performed each significant step of the process and, where appropriate, the name of any person who checked these operations.
- iv. Records of checks for identity and conformity with the packaging instructions, including the results of in-process controls.
- v. Records of checks that the equipment and workstation are clear of previous products, documents or materials not required for the planned packaging operations (line clearance), that equipment is clean and suitable for use, and that any electronic code readers, label counters or similar devices are functioning as expected.
- vi. Details of the packaging operations carried out, including references to equipment and the packaging lines used.
- vii. Whenever possible, samples of printed packaging materials used, including specimens of the batch coding, expiry dating and any additional overprinting.
- viii. Notes on any special problems or unusual events including details, with signed

- 270 authorisation for any deviation from the Packaging Instructions.
- 271 ix. The quantities and reference number or identification of all printed packaging mate-
272 rials and bulk product issued, used, destroyed or returned to stock and the quantities
273 of obtained product, to provide for an adequate reconciliation. Where there are vali-
274 dated electronic controls in place during packaging there may be justification for not
275 including this information
- 276 x. Approval by the person responsible for the packaging operations.

277 **Receipt**

- 278 4.39. There should be written procedures and records for the receipt of each delivery of each starting
279 material, (including bulk, intermediate or finished goods), primary, secondary and printed
280 packaging materials and QC-samples. The records of the receipts should include:
- 281 i. The name of the material on the delivery notes and the containers.
- 282 ii. The "in-house" name and/or code of material (if different from a).
- 283 iii. Date of receipt.
- 284 iv. Supplier's name and, manufacturer's name.
- 285 v. Manufacturer's batch or reference number.
- 286 vi. Total quantity and number of containers received.
- 287 vii. The batch number assigned after receipt.
- 288 viii. Any relevant comment.
- 289 ix. If applicable, proof of verification that temperature during transportation were within
290 the approved limit.
- 291 4.40. There should be written procedures for the internal labelling, quarantine and storage of start-
292 ing materials, packaging materials, QC samples and other materials, as appropriate.

293 **Sampling**

- 294 4.41. There should be written procedures for sampling, which include the methods and equipment
295 to be used, the amounts to be taken and any precautions to be observed to avoid contamination
296 of the material or any deterioration in its quality (reference to EU GMP Guideline Volume 4,
297 Chapter 6 "Quality Control").

298 **Testing**

- 299 4.42. There should be written procedures for testing materials and products at different stages of
300 manufacture, describing the methods and equipment to be used. The tests performed should
301 be recorded (reference to EU GMP Guideline Volume 4, Chapter 6 "Quality Control").

302 **Other**

- 303 4.43. Written release and rejection procedures should be available for materials and products, and
304 in particular for the certification for sale of the finished product by the Qualified Person(s).
305 All records should be available to the Qualified Person at the time of the release decision. A
306 system should be in place to indicate special observations and any changes to critical data.
- 307 4.44. Records should be maintained for the distribution of each batch of a product in order to facil-
308 itate recall of any batch, if necessary.
- 309 4.45. There should be written policies, procedures, protocols, reports and the associated records of
310 actions taken, or conclusions reached, where appropriate, for GMP relevant actions, including

but not limited to the following examples:

- i. Validation and qualification of processes, equipment and systems.
 - ii. Equipment assembly and calibration.
 - iii. Data integrity.
 - iv. Technology transfer.
 - v. Maintenance, cleaning and sanitation.
 - vi. Personnel matters including signature lists, training in GMP and technical matters, clothing and hygiene and verification of the effectiveness of training.
 - vii. Environmental monitoring.
 - viii. Pest control.
 - ix. Complaints.
 - x. Recalls.
 - xi. Returns.
 - xii. Change control.
 - xiii. Investigations into deviations.
 - xiv. non-conformances e.g. out of specifications.
 - xv. Internal quality/GMP compliance audits.
 - xvi. Summaries of records where appropriate (e.g. product quality review).
 - xvii. Supplier audits.
- 4.46. Clear operating procedures should be available for major items of manufacturing and test equipment.
- 4.47. Logbooks should be kept for major or critical analytical testing, production equipment, and areas where product has been processed or handled. They should be used to record in chronological order, as appropriate, any use of the area, equipment/method, calibrations, maintenance, cleaning or repair operations, including the dates and identity of people who carried these operations out.
- 4.48. An inventory of documents within the pharmaceutical quality system should be maintained.

GENERATION AND CONTROL OF DOCUMENTATION

- 4.49. All types of documents (instructions and/or records) should be defined and adhered to regardless of the documentation technology, hybrid solution or service. The technology, hybrid solution or provided service needs to be understood regardless of complexity, should be adequately documented, and validated with risk-based controls in place. Relationships and control measures for master documents, official copies, data handling and records need to be defined for both hybrid and homogenous systems regardless of the type of service. Appropriate controls for documents should be implemented to ensure the completeness, integrity and legibility of the records throughout the lifecycle.
- 4.50. Documents should be designed, prepared, reviewed, and distributed in a controlled manner. They should comply with the relevant parts of Product Specification Files, Manufacturing and Marketing Authorisation dossiers, or dossiers of Investigational Medicinal Products, as appropriate. The reproduction of working documents from master documents should not allow any error or alteration to be introduced through

the reproduction process.

4.51. Documents should be regularly reviewed and kept up to date. Documents should be approved, signed, and dated by appropriate and authorised personnel. Documents should have unambiguous contents and be uniquely identifiable. The effective date should be defined.

4.52. Documents containing instructions should be laid out in an orderly fashion and be easy to review. The style and language of documents should fit with their intended use. Standard operating procedures, work instructions and methods should be written in an imperative mandatory style by using predefined format. Data entry formats for completion of documents should be clearly defined. Written instructions may be supported with pictures, photos or videos. The documents containing the instructions should be easily accessible at the place where the described activities are carried out.

4.53. Instructions and procedures within the Quality Management System should be regularly reviewed and kept updated.

4.54. The issuance, revision, superseding and withdrawal of all documents should be controlled with maintenance of revision histories.

4.55. Hand-written instructions are discouraged. Where documents require the manual entry of data, sufficient space should be provided for such entries to ensure adequately clear and legible manual recording.

GOOD DOCUMENTATION PRACTICE

4.56. Good Documentation practices are key to ensuring data integrity, and a fundamental part of a well-designed pharmaceutical quality system.

4.57. Procedures outlining good documentation practices and arrangements for document control should be available within the pharmaceutical quality system. Good documentation practices should be implemented and enforced to ensure data integrity.

4.58. Data entries should be accurate, and made in clear, legible, indelible way. Recorded media should be durable throughout the retention period. If this is not feasible, then true copies should be generated. For this case a documented system should be in place to verify and record the integrity of the copy.

4.59. Records should be made or completed at the time each action is taken and in such a way that all GMP activities are traceable. It should be possible to identify the individual or the system that performed the task and when the task was performed.

4.60. Any alteration made to the entry on a document should be signed by the individual who made the change and dated; the alteration should permit the reading of the original information. Where appropriate, the reason for the alteration should be recorded.

4.61. Records need to be a truthful and consistent representation of facts to ensure the accuracy of information, including data that is used to make critical decisions about the quality of products.

4.62. Specific controls should be implemented to ensure the integrity of raw data and results recorded on paper. These may include, but are not limited to:

i. control over the issuance and use of loose paper sheets (blank forms) at the time of recording data.

ii. control over the issuance of bound, paginated notebooks.

iii. control over the issuance and reconciliation of sequentially numbered copies of

blank forms with authenticity controls.

- iv. Control that raw data is contemporaneously and accurately recorded by permanent means.

4.63. Basic data integrity principles (table 1) applicable to both paper and electronic systems (i.e. ALCOA ++):

Table 1: Data integrity principles

Attributes	Requirement
Attributable	It should be possible to identify the individual or computerised system that performed a recorded task and when the task was performed. This also applies to any changes made to records, such as corrections, deletions, and changes where it is important to know who made a change, when, and why.
Legible	All records should be legible – the information should be readable and unambiguous in order to be understandable and of use. This applies to all information that would be required to be considered complete, including all original records or entries. Where the ‘dynamic’ nature of electronic data (the ability to search, query, trend, etc.) is important to the content and meaning of the record, the ability to interact with the data using a suitable application is important to the ‘availability’ of the record.
Contemporaneous	The evidence of actions, events or decisions should be recorded as they take place. This documentation should serve as an accurate attestation of what was done, or what was decided and why, i.e. what influenced the decision at that time.
Original	The original record can be described as the first capture of information, whether recorded on paper (static) or electronically (usually dynamic, depending on the complexity of the system). Information that is originally captured in a dynamic state should remain available in that state.
Accurate	Records need to be a truthful representation of facts to be accurate. Ensuring records are accurate is achieved through many elements of a robust pharmaceutical quality system. This can be comprised of: • equipment related factors such as qualification, calibration, maintenance, and computer validation. • policies and procedures to control actions and behaviours, including data review procedures to verify adherence to procedural requirements.

	• deviation management including root cause analysis, impact assessments and CAPA. • trained and qualified personnel who understand the importance of following established procedures and documenting their actions and decisions. Together, these elements aim to ensure the accuracy of information, including scientific data that is used to make critical decisions about the quality of products.
Complete	All information that would be critical to recreating an event is important when trying to understand the event. It is important that information is not lost or deleted. The level of detail required for an information set to be considered complete would depend on the criticality of the information. A complete record of data generated electronically includes relevant metadata.
Consistent	Information should be created, processed, and stored in a logical manner that has a defined consistency. This includes policies or procedures that help control or standardize data (e.g. chronological sequencing, date formats, units of measurement, approaches to rounding, significant digits, etc.).
Enduring	Records should be kept in a manner such that they exist for the entire period during which they might be needed. This means they need to remain intact and accessible as an indelible/durable record throughout the record retention period.
Available	Records should be available for review at any time during the required retention period, accessible in a readable format to all applicable personnel who are responsible for their review whether for routine release decisions, investigations, trending, annual reports, audits or inspections.
Traceable	Traceability is the ability to trace the history, modification or location of data by means of recorded identifications.

404

405 SIGNATURES IN GMP RELEVANT DOCUMENTATION

406 4.64. Signatures are essential for ensuring accountability for activities in a GMP environ-
407 ment at the time points the signatures are executed.

408 4.65. A signature represents the legally binding will of the signatory. The signatory should
409 sign with date and time. If signatures by initials are in use a procedure defining abbrevi-
410 ated signatures should be in place.

- 4.66. The identification of the signatory should be possible. Data or documents which are associated with the signature should be clearly identified. The meaning of the signature (such as review, approval, responsibility, or authorship) associated with the signature should be clear.
- 4.67. The regulated user should establish a signature policy to ensure the adequate application of signatures. Personnel authorised to sign should be clearly identified by name and bound by name to the signature policy.
- 4.68. The regulated user should have identified those records which require a legally binding signature.
- 4.69. Signatures should be indisputable and traceable to the signatory and the signed document or record, regardless, if a signature is applied on paper or electronically.
- 4.70. If records exist electronically such records should be signed electronically. The use of a hybrid system should be avoided. If signatures exist parallel in paper and electronically (e.g. in hybrid systems), the regulatory relevant signature should be defined by the regulated user.
- 4.71. The signatory should be qualified and authorised to perform the relevant tasks or reviews.
- 4.72. The regulated user should define the signatory's role and responsibility in the signation process.
- 4.73. The regulated user should ensure that the signatory's role and qualification is consistent with the meaning (intent) of a signature.
- 4.74. To ensure the integrity of signatures during the whole life cycle of data the regulated user should establish the management and control of signatures as an element of a data governance system.
- 4.75. The data or documents which the signature is relevant for should fulfil the ALCOA++ principles.

RETENTION OF DOCUMENTS

- 4.76. It should be clearly defined which record is related to each activity and where this record is located, regardless of the technology, hybrid solution or service used. Risk-based control methods should be in place to ensure the integrity of the record throughout the lifecycle. The control measures should be covered by the validation scope. In case of electronic recording such measures should include back-up, restore and archiving procedures as well as physical and logical controls. If the regulated user relies on hosted services, it is the responsibility of the regulated user to understand, approve and justify the control measures of the hosted service provider based on a service level agreement. Records should be available for review at any time during the required retention period, accessible in a human readable format to all applicable personnel.
- 4.77. Specific requirements apply to batch documentation which must be kept for one year after expiry of the batch to which it relates or at least five years after certification of the batch by the Qualified Person, whichever is the longer. For investigational medicinal products, the batch documentation must be kept for at least five years after the completion or formal discontinuation of the last clinical trial in which the batch was used. Other requirements for retention of documentation may be described in legislation in relation to specific types of products (e.g. Advanced Therapy Medicinal

Products and products derived from human blood or human plasma) and specify that longer retention periods be applied to certain documents.

- 4.78. For other types of documentation, the retention period will depend on the business activity which the documentation supports. Critical documentation, including raw data (for example relating to validation or stability), which supports information in the Marketing Authorisation should be retained whilst the authorization remains in force. It may be considered acceptable to retire certain documentation (e.g. raw data supporting validation reports or stability reports) where the data has been superseded by a full set of new data. Justification for this should be documented and should consider the requirements for retention of batch documentation; for example, in the case of process validation data, the accompanying raw data should be retained for a period at least as long as the records for all batches whose release has been supported on the basis of that validation exercise.
- 4.79. A documented process for the disposal of records should be in place to ensure that the correct original records or true copies are disposed only after the defined retention period. Measures should be in place to reduce the risk of deleting the wrong documents. The access rights allowing disposal of records should be controlled.

DATA INTEGRITY IN DOCUMENTATION

- 4.80. The method of documentation should be integrated in the regulated user's pharmaceutical quality system. Documents or records should be controlled in a risk-based approach regardless of whether located in-house or in the form of hosted services. The regulated user should apply the principles of data integrity, data criticality and data risk within a data governance system and should consider the complete lifecycle of data. The data governance system should be an element of the pharmaceutical quality system. The ownership of data and the responsibility for data integrity should be defined.
- 4.81. Risk based control measures should be commensurate with the type and the complexity of a system. The pharmaceutical quality system should interface with independent review practices to detect risks to data integrity. Risks from human factors should be considered for effectively ensuring data integrity Risk-reducing measures such as second person oversight, verification and checks should be implemented where appropriate and in the appropriate time to ensure critical process and testing steps are accurately and contemporaneously recorded.

HYBRID SYSTEMS

- 4.82. They should be clearly defined and identified, and each contributing element of the system validated and controlled according to risk management principles.
- 4.83. A detailed description of the entire system should be available. The description should outline all major components, their functions, and interactions with each other as well as control for data management and data integrity. Procedures and records should be available to manage and appropriately control the interface between manual and computerised systems.
- 4.84. Appropriate quality risk management principles should be followed when assessing, defining, and demonstrating the effectiveness of control measures applied to the system.
- 4.85. Procedures should be in place to manage the review of data generated by hybrid

501 systems which clearly outline the process for the evaluation, approval and archiving
502 of electronic and paper-based data.

503 GLOSSARY

504 ALCOA++ An acronym for “attributable, legible, contemporaneous, original and accu-
505 rate”, which puts additional emphasis on the attributes complete, consistent,
506 enduring, available and traceable – implicit basic ALCOA principles.

507 Archiving Long term, or permanent retention of completed documentation and relevant
508 metadata in its final form for the purposes of reconstruction of a process or
509 activity.

510 Automated script A piece of code used to automate repetitive processes

511 Data The contents of the record. Data may be defined as measurable or descriptive
512 attribute of a physical entity, process, or event.

513 Data governance The total sum of arrangements to ensure that data, irrespective of the format in
514 which it is generated, recorded, processed, retained and used, will be attribut-
515 able, legible, contemporaneous, original, accurate, complete, consistent, en-
516 during, and available throughout the data lifecycle.

517 Data integrity Data integrity refers to the completeness, consistency, and accuracy of data.
518 Complete, consistent, and accurate data should be attributable, legible, con-
519 temporaneously recorded, original or a true copy, accurate and traceable (AL-
520 COA++).

521 Data lifecycle All processes related to the creating, recording, processing, reviewing, chang-
522 ing, analysing, reporting, transferring, storing, migrating, archiving, retrieving,
523 and deleting of data.

524 Data management The set of all methodological, conceptual, organisational and technical
525 measures and procedures for handling data with the aim of incorporating it into
526 business processes.

527 Data risk The combination of the probability of occurrence of harm and the severity of
528 that harm related to data (incompleteness, alterations or loss which compro-
529 mise the integrity of data).

530 Data criticality The degree of influence that data have on product safety as well as the regula-
531 tory compliance of processes, decisions and product quality.

532 Data Risk Management An activity to be applied throughout the lifecycle of data considering the need
533 to ensure data integrity. Risk management consists of risk identification, risk
534 assessment, risk mitigation and risk control. Risk management should link to
535 other relevant procedures (e.g. configuration and change management, man-
536 agement processes for data, business risks, etc.).

537 Data Risk Assessment The process of evaluating the risks associated with the regulated user’s data. It
538 ensures an efficient and effective approach to data integrity by considering the
539 vulnerability of data to involuntary or deliberate alteration resulting in risk-
540 based control measures.

541	Data ownership	The allocation of responsibilities for control of data to a specific process
542		owner. Companies should implement systems to ensure that responsibilities for
543		systems and their data are appropriately allocated, and responsibilities under-
544		taken.
545	Data quality	The degree to which a set of inherent characteristics (quality dimensions) of
546		data fulfils requirements.
547		Data should be fit for use in their intended operational, decision-making, and
548		other roles and should exhibit conformance to regulatory standards that have
549		been set, so that fitness for use is achieved.
550	Document	A formatted compilation of data. Operations and activities that are memorial-
551		ized in (electronic) records may consist of one or more documents that describe
552		the activity in a moment of time.
553	Electronic record	Any combination of text, graphics, data, audio, pictorial, or other information
554		representation in digital form that is created, modified, maintained, archived,
555		retrieved, or distributed by a computer system.
556	Hybrid system	A combination of paper based and electronic means.
557	Homogenous systems	A system that is either paper or electronically based on-premises or a cloud
558		service.
559	Meta data	Describe the attributes of data and provides context and meaning. Metadata is
560		any information used for the identification, description, and relationships of
561		electronic records or their elements. Metadata gives data meaning, provides
562		context, defines structure, and enables retrievability across systems, and usa-
563		bility, authenticity, and auditability across time.
564	Raw Data	Raw data is defined as the original record (data) which can be described as the
565		first capture of stored information, whether recorded on paper or electronically.
566		Information that is originally captured in a dynamic state should remain avail-
567		able in that state.
568	Record	Memorializes, or makes information permanent about, an action, activity and
569		event that caused its creation.
570	Regulated user	Marketing Authorisation Holder, Manufacturers, control laboratories, import-
571		ers, and wholesale distributors (if the wholesale distributor holds a manufac-
572		turing license).
573	Risk based approach for data integrity	
574		A process to define critical data, documents and the actions used to monitor
575		activities like capturing, derivation, migration, storage, communication and ar-
576		chiving to ensure that data and documents remain in a state of control through-
577		out the entire lifecycle and to maintain its integrity.
578	Specification	A list of tests, references to analytical procedures, and appropriate acceptance
579		criteria that are numerical limits, ranges, or other criteria for the test described.
580		It establishes the set of criteria to which a material should conform to be con-
581		sidered acceptable for its intended use. "Conformance to specification" means
582		that the material, when tested according to the listed analytical procedures, will
583		meet the listed acceptance criteria.

584	True copy	An exact copy of original documentation that preserves the same content,
585		meaning and attributes of the original. The term “true copy” is synonymous
586		with “certified” or “verified copy”.
587	Type of service	On-premises IT service or outsourced hosted (cloud) IT service
588	Verified copy	Refer to definition of true copy
589	Data Risk Assessment	The process of evaluating the risks associated with the regulated user’s data.